

Ryan Herrmann

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EXPERIENCE

Teaching Assistant

Aug. 2025 – Present

CU Boulder

Boulder, CO

- Instruct electrical circuits laboratories for 200 students each semester, supporting hands-on learning and technical problem solving
- Teach foundational circuit laws, analysis techniques, and basic circuit design, reinforcing lecture concepts through applied experiments
- Demonstrate and supervise use of laboratory equipment including function generators, op-amps, semiconductors, servo motors, and sensors

Field Service Engineer

Apr. 2025 – Aug. 2025

IQVIA

Remote / Nationwide

- Collaborated with Medtronic to upgrade over 50 colonoscopy and endoscopy imaging machines with increased polyp detection and accuracy
- Deployed software updates to clinical equipment at 23 facilities nationwide, ensuring uniform functionality across sites
- Managed communication between on-site, sales, and internal representatives

EDUCATION

University of Colorado Boulder

Boulder, CO

Master of Science, Mechanical Engineering (GPA: 3.85)

Aug. 2025 – May 2027

- **Coursework:** Fluid Dynamics, CFD, Design

California Polytechnic State University

San Luis Obispo, CA

Bachelor of Science, Biomedical Engineering, Bioinstrumentation Emphasis (GPA: 3.48)

Sep. 2020 – Jun. 2024

- **Achievements:** President's List 2024
- **Coursework:** Biomechanics, Biomaterials, Biosensors and Circuits, Biopotential Instrumentation, Neural Systems Simulation and Modeling, Fluid Dynamics, Thermodynamics, Linear Analysis II

PROJECTS

Trailhead Safety Indicator | *SPICE, Circuit Design, Power Systems*

Jan. 2026 – May. 2026

- Led design of a passive PMOS power latch circuit using an asymmetric RC network to solve Arduino cold-start power sequencing, verified through SPICE simulation
- Designed a discrete two-stage PNP/NPN transistor level shifter to interface a 5V RTC alarm signal with a 7.4V PMOS gate, maintaining full domain isolation between power rails
- Achieved 25-day standby battery life against a 20-day design specification through system power architecture design, reducing standby current draw to ~5 mA by fully cutting power to all non-essential components during sleep

Water Hammer CFD Study | *CFD, Python, DOLFINx, FEA*

Jan. 2026 – May 2026

- Developed a 2D axisymmetric monolithic FSI solver in DOLFINx using Nitsche interface coupling, $k-\epsilon$ turbulence closure, and a barotropic equation of state to physically capture Joukowski pressure wave propagation
- Benchmarked Crank-Nicolson vs. BDF2 time integration schemes across 9 pipe material and working fluid combinations, validating simulated peak pressures against analytical Joukowski theory
- Co-authored a technical report documenting FSI discretization limits, scheme-dependent amplitude behavior, and compliant-pipe breakdown regimes for steel, copper, and HDPE pipes

Biomedical Senior Design | *SolidWorks, 3D Printing, Mechanical Design*

Sep. 2023 – Mar. 2024

- Designed and prototyped a wrist-worn tremor damping device for individuals with essential tremor or Parkinson's disease, targeting the 5–7 Hz pathological tremor frequency range
- Developed a spring and air damper mechanism housed in a 3D printed enclosure, iterating through 6 design revisions to refine damping performance
- Co-authored a 24-page technical report documenting design requirements, iterative development, and final prototype evaluation

TECHNICAL SKILLS

Programming & Data Analysis: MATLAB, Python, Git, Java, JMP

Design & Simulation: SolidWorks, COMSOL, LTSpice, MINDI, KiCad, DOLFINx, MPLAB

Tools & Environments: Linux, WSL, tmux, Remote Terminal Workflows